

## RCC Structural Drawing Compliance Report - NH OFFICE FOUNDATIONS

**Site Location:** Mangalore

**Reference Codes:** IS 456:2000, SP 34

### Introduction

This report provides a comprehensive compliance analysis of the provided RCC structural drawing for "FOUNDATIONS" of the "NH OFFICE" building. The analysis integrates previous findings with new user-provided information (Site: Mangalore) and cross-references all aspects against IS 456:2000 and SP 34. Where information was previously missing, limiting values as per the codes have been assumed for compliance checks, as per user instruction.

### Step 0: Initial Document Check

- **0.1:** The document is an RCC structural drawing of "FOUNDATIONS" only.
- **0.2:** The site location is "NH OFFICE" in Mangalore.
- **0.3:** All compliance checks are based on IS 456:2000 and SP 34.

### Step 1: Locate the "NOTES" Section

- **Status:** Compliant. The "NOTES" section is clearly present on the drawing.

### Step 2 & 3: Extract and Verify Design Parameters from "NOTES"

| Criteria                     | Extracted Value                                                                                                                                                                           | Compliance Check                                                                                                                                                                                                                           | Status    | IS Code Reference        |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------|
| ---                          | ---                                                                                                                                                                                       | ---                                                                                                                                                                                                                                        | ---       | ---                      |
| <b>1. Grade of Concrete</b>  | M25 for all R.C.C works.                                                                                                                                                                  | For Mangalore, which is a coastal area, the exposure condition is typically 'Severe' as per IS 456:2000, Table 3. For 'Severe' exposure, the minimum grade of concrete for RCC is M25 (IS 456:2000, Table 5). Therefore, M25 is compliant. | Compliant | IS 456:2000, Table 3 & 5 |
| <b>2. Reinforcement Bars</b> | HYSD TMT BARS of Grade Fe 500 conforming to IS 1786-1985. Shear reinforcement and stirrups are indicated by Y8@150 C/C and Y8@200 C/C in the lift pit section and Y8@200 for column ties. | Fe 500 is a standard and compliant grade for HYSD bars (SP 34, Clause 12, Table for IS 1786-1985). The specified bar sizes and spacing for shear reinforcement and ties are noted.                                                         | Compliant | SP 34, Clause 12         |

|                                               |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |                            |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------|
| <b>3. Lap Length</b>                          | 50 times dia of the bar and to be staggered such that not more than 50% of the bars are lapped at a section. | This is compliant with general practice and IS 456:2000 requirements for lap length, which typically ranges from 45d to 60d depending on stress and concrete grade. The staggering requirement is also compliant with good detailing practice to avoid planes of weakness.                                                                                                                                                                                                                                                                                                                                      | Compliant     | IS 456:2000, Clause 26.2.5 |
| <b>4. Clear Cover</b>                         | Footing/Wall: 50 mm, Columns: 40 mm, Slab: 20 mm, Beam: 25 mm.                                               | For 'Severe' exposure (Mangalore), minimum clear cover for footings is 50mm (IS 456:2000, Table 16A). The provision of PCC below footing further protects the reinforcement. Column cover of 40mm is compliant for 'Severe' exposure (IS 456:2000, Table 16A). Slab cover of 20mm and Beam cover of 25mm are compliant for 'Moderate' exposure, but for 'Severe' exposure, minimum 30mm is recommended for beams and 25mm for slabs (IS 456:2000, Table 16A). However, for slabs, 20mm is often accepted for mild exposure. Given the 'Severe' exposure, the beam and slab covers are marginally non-compliant. | Non-Compliant | IS 456:2000, Table 16A     |
| <b>5. Development Length (Ld)</b>             | 50 times the dia. of the bar.                                                                                | This value is a common and generally compliant practice for development length, often used as a conservative estimate. The actual development length depends on the grade of concrete, grade of steel, and bond stress (IS 456:2000, Clause 26.2.1).                                                                                                                                                                                                                                                                                                                                                            | Compliant     | IS 456:2000, Clause 26.2.1 |
| <b>6. Safe Bearing Capacity (SBC) of soil</b> | 12.5 T/m <sup>2</sup> . PCC is mentioned as "5" THK. PCC" in Section X1.                                     | SBC is provided. The provision of PCC below footing is good practice and compliant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Compliant     | -                          |
| <b>7. Seismic Zone and Wind Load</b>          | Structure is designed for Seismic Zone III and Wind Forces as per IS 1893:2016 & IS 875:2015 respectively.   | This information is mandatory and provided, indicating consideration of lateral loads.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Compliant     | -                          |

|                                                 |                                                                                                                                                                      |                                                                                                                                                                     |                |                  |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------|
| <b>8. Building Limitations</b>                  | Structure is designed for Ground +2 Stories only.                                                                                                                    | Building limitations are mentioned, which is important for design assumptions.                                                                                      | Compliant      | -                |
| <b>9. Structure's Purpose</b>                   | NH OFFICE (Foundation Layout).                                                                                                                                       | The purpose is mentioned as "NH OFFICE" and the drawing title is "FOUNDATION LAYOUT".                                                                               | Compliant      | -                |
| <b>10. Floor Heights</b>                        | First Floor: 3.6M, Second Floor: 3.6M, Terrace: 3.6M (from plinth beam).                                                                                             | Floor heights are mentioned, providing necessary dimensional data.                                                                                                  | Compliant      | -                |
| <b>11. Schedule of Footings</b>                 | "SCHEDULE OF FOOTINGS" table is present and consistent with the drawing.                                                                                             | The table is present and appears consistent, which is essential for construction.                                                                                   | Compliant      | -                |
| <b>12. Footing Type</b>                         | Isolated footings (F1) and Raft footings (RF1, RF2) are shown.                                                                                                       | Isolated footings are suitable for a G+2 storey building. The presence of raft footings suggests varying soil conditions or load requirements, which is acceptable. | Compliant      | SP 34, Section 6 |
| <b>13. Reinforcement in High-Rise Buildings</b> | The building is G+2 stories, which is not considered high-rise. Top reinforcement is shown for raft footings (RF1, RF2).                                             | Not Applicable as the building is not high-rise. Top reinforcement in raft footings is standard practice.                                                           | Not Applicable | -                |
| <b>14. Raft Foundation Reinforcement</b>        | RF1: Top Y12@125, Bottom Y16@150. RF2: Top Y12@125, Bottom Y16@125.                                                                                                  | Both top and bottom reinforcement are specified for raft foundations, which is compliant with design requirements for bending moments.                              | Compliant      | SP 34, Section 6 |
| <b>15. Lift Design</b>                          | A lift pit is shown with a depth of "1500 MIN. OR AS PER ARCH." and reinforcement details. Bending of steel reinforcement at the footing base is shown in Section X. | Lift pit depth is specified as minimum 1500mm. Bending details are shown, which is good practice for structural integrity.                                          | Compliant      | -                |
| <b>16. Soil Improvement</b>                     | "90 TO 40MM SIZE BOULDERS COMPACTED WITH SAND/QUARRY DUST" is mentioned below the footing in Section X.                                                              | Soil improvement details are provided, indicating consideration for foundation stability.                                                                           | Compliant      | -                |

|                                                |                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                             |                     |                                        |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------------------------|
| <b>17. Column Ties</b>                         | Column ties are shown in the column schedules (e.g., Y8@200). They appear continuous in the typical column sections.                                                                                         | Column ties are specified and appear continuous, which is essential for confining longitudinal reinforcement and resisting shear.                                                                                                                                                                                                                                                                                           | Compliant           | IS 456:2000, Clause 26.5.3.2           |
| <b>18. Plan of Ties</b>                        | A separate plan for ties is not explicitly present, but tie details are provided within the column schedules.                                                                                                | While a separate plan is not present, the details being provided in the column schedules is acceptable for a G+2 structure.                                                                                                                                                                                                                                                                                                 | Compliant           | SP 34, Section 7                       |
| <b>19. Outer Ties Check</b>                    | Column schedules specify ties (e.g., Y8@200). Percentage of steel in columns is not explicitly stated in the notes but can be calculated from the column schedules. Footing/slab reinforcement is specified. | Assuming limiting values as per user input: Minimum percentage of steel in columns is 0.8% (IS 456:2000, Clause 26.5.3.1). Minimum percentage of steel in footings/slabs is 0.12% for Fe 500 (IS 456:2000, Clause 26.5.2.1). These values are not explicitly stated in the notes, requiring calculation from detailed drawings. Without specific dimensions and bar counts, direct verification from notes is not possible. | Missing Information | IS 456:2000, Clause 26.5.3.1, 26.5.2.1 |
| <b>20. Cross-Section Area</b>                  | Not explicitly mentioned whether gross cross-section area is used for columns/footings and effective area for slabs.                                                                                         | This information is not specified in the notes. For columns, gross area is typically used for axial load calculations, and for slabs, effective depth is used for flexural design. This is a design assumption that should ideally be stated.                                                                                                                                                                               | Missing Information | -                                      |
| <b>21. Steel Curtailment</b>                   | The building is G+2 stories. Steel curtailment is not explicitly mentioned in the notes.                                                                                                                     | For a G+2 building, curtailment might not be as critical as for high-rise, but it's good practice to mention general principles or refer to relevant clauses. It's not explicitly detailed in the notes.                                                                                                                                                                                                                    | Missing Information | IS 456:2000, Clause 26.2.3             |
| <b>22. Maximum Steel Percentage in Columns</b> | Not explicitly mentioned in the notes.                                                                                                                                                                       | Assuming limiting values as per user input: Maximum percentage of steel in columns is 6% (IS 456:2000, Clause 26.5.3.1). This value is not stated in the notes.                                                                                                                                                                                                                                                             | Missing Information | IS 456:2000, Clause 26.5.3.1           |

### **Step 5: Report Missing or Wrong Information**

**Clear Cover for Beams and Slabs:** For 'Severe' exposure conditions in Mangalore, the specified clear cover for beams (25mm) and slabs (20mm) is less than the minimum recommended by IS 456:2000, Table 16A (30mm for beams, 25mm for slabs).

\* **Recommendation:** Increase clear cover for beams to at least 30mm and for slabs to at least 25mm, or provide justification for the current values based on specific design considerations (e.g., use of higher grade concrete, special coatings).

**Outer Ties Check (Percentage of Steel):** The minimum percentage of steel in columns (0.8%) and footings/slabs (0.12% for Fe 500) is not explicitly stated in the general notes. While these can be calculated from the detailed drawings, their explicit mention in general notes would enhance clarity.

\* **Recommendation:** Add a note specifying the minimum percentage of reinforcement used for columns and footings/slabs, or confirm that these percentages are met through detailed calculations.

**Cross-Section Area Usage:** It is not explicitly mentioned whether gross cross-section area is used for columns/footings and effective area for slabs in design calculations.

\* **Recommendation:** Add a note clarifying the basis for cross-section area usage in design.

**Steel Curtailment:** General principles or specific clauses regarding steel curtailment are not explicitly mentioned in the notes.

\* **Recommendation:** Include a general note on steel curtailment practices, referencing IS 456:2000, Clause 26.2.3, or indicating that curtailment is as per detailed drawings.

**Maximum Steel Percentage in Columns:** The maximum percentage of steel in columns (6%) is not explicitly mentioned in the notes.

\* **Recommendation:** Add a note specifying the maximum percentage of reinforcement used for columns, referencing IS 456:2000, Clause 26.5.3.1.

### **Summary of Compliance**

- **Total Criteria Evaluated:** 22
- **Compliant Items:** 16
- **Non-Compliant Items:** 1 (Clear Cover for Beams and Slabs)
- **Missing Information Items:** 4
- **Cannot Verify Items:** 0 (Resolved by user input and limiting values)
- **Not Applicable Items:** 1

#### **Overall Verdict:**

The structural drawing for the foundations of "NH OFFICE" in Mangalore is largely compliant with IS 456:2000 and SP 34, especially regarding fundamental design parameters and material specifications. The user's input regarding the site location (Mangalore) allowed for a definitive check on the concrete grade and clear cover requirements based on exposure conditions.

However, a minor non-compliance was identified regarding the clear cover for beams and slabs under 'Severe' exposure conditions. Additionally, several items related to explicit statements of design assumptions (e.g., percentage of steel, cross-section area usage, steel curtailment) are still missing from the general notes. While these might be implicitly covered in the detailed design, their explicit mention would significantly improve the clarity and completeness of the drawing notes.